Adaptive deep brain stimulation in Parkinson's disease

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With the introduction of adaptive deep brain stimulation (aDBS) for Parkinson's disease, new questions emerge regarding who, why, and how to treat. This paper outlines the pathophysiological rationale for aDBS, which provides real-time modulation of the stimulation amplitude based on subthalamic beta (range 13–30 Hz) activity and related physiomarkers. We review clinical evidence comparing aDBS with conventional DBS in terms of motor improvement, side-effect reduction, energy efficiency, and technical developments, including sensing-enabled device characteristics, stimulation algorithms, and potential clinical indications. We also discuss limitations, such as physiomarker variability, signal artifacts, and the absence of standardised programming protocols. Finally, we explore the readiness for clinical implementation and future directions, and estimate the scope of eligible patients. In our view, aDBS marks a fundamental change in approach from fixed stimulation towards physiomarker-guided neuromodulation. This evolution necessitates new infrastructure, clinician training, and real-world studies, but holds promise for more personalised and responsive treatment.

Lancet 2026; 407: 1191–204

Published **Online** February 18, 2026
https://doi.org/10.1016/S0140-6736(25)02274-3

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Introduction

Parkinson's disease is a neurodegenerative disorder that is rising in prevalence,[1] characterised by bradykinesia with tremor or rigidity due to a progressive loss of dopaminergic neurons in the substantia nigra.[2] In addition, Parkinson's disease involves a wide range of non-motor symptoms, such as depression, anxiety, sleep problems, hyposmia, cognitive dysfunction, and constipation. These symptoms can be attributed to both dopaminergic and non-dopaminergic neurodegeneration.[3] Some, such as hyposmia and rapid eye movement sleep behaviour disorder, can precede motor onset, whereas others, including Parkinson's disease dementia and hallucinations, typically develop later.[4] The progression of these symptoms and the disease is linked to the spread of Lewy body pathology throughout the nervous system, and varies substantially between patients.[5,6]

No curative or disease-modifying therapy is currently available for Parkinson's disease. Although numerous trials are in progress, treatment is symptomatic and typically begins with dopaminergic medication.[7,8] Although this treatment results in a substantial initial improvement, most patients develop motor complications (response fluctuations) within 9 years of diagnosis—the medication either induces disabling involuntary movements (dyskinesias) or loses efficacy.[9,10] The result is often unpredictable fluctuations between OFF states with severe motor symptoms and ON states with troublesome dyskinesias, despite complex medication regimens.

Deep brain stimulation (DBS), a neurosurgical treatment that delivers continuous electrical stimulation to targeted deep-brain structures via implanted electrodes, is a therapeutic option in the advanced stages of Parkinson's disease.[11] It requires the implantation of one or more electrodes, with stereotactic surgery, in deep brain structures, such as the subthalamic nucleus (STN) and the internal segment of the globus pallidus (GPi). Beginning at around 2 weeks after surgery, the electrodes deliver electrical pulses continuously throughout the day, typically in the milliamp range and at a frequency of around 130 Hz. Although the precise mechanism of DBS in Parkinson's disease is not yet fully understood and likely involves a complex interplay between inhibitory and excitatory effects—resulting in both short-term and long-term modulation of activity in the basal ganglia–thalamo–cortical network[12]—the prevailing theory suggests that the high-frequency stimulation disrupts pathological oscillatory neural activity in the beta frequency range (13–30 Hz), restoring balance between the direct (facilitating movement) and indirect (inhibiting movement) cortico–basal ganglia pathways.[13] Extensive clinical trials have shown that DBS of the STN or GPi substantially improves motor symptoms such as bradykinesia and tremor, increases time spent in the ON state, allows for a reduction of dopaminergic medication, and enhances quality of life.[14] Additionally, DBS has proven effective even in patients with early motor complications[15] and the resulting motor improvements appear to be sustained for more than a decade.[16]

Despite its benefits, DBS has several limitations. Beyond surgical risks and perioperative complications, its effectiveness is often constrained by stimulation-induced side-effects. One of the most documented effects, particularly in STN DBS, is stimulation-induced dysarthria.[17] This side-effect often limits the ability to increase the stimulation intensity to an amount required

Search strategy and selection criteria

We obtained the referenced literature for this publication through searches of PubMed from database inception to Feb 13, 2025, without restrictions on language. Both medical subject headings and free-text terms were used to identify relevant articles. The terms "Parkinson's disease", "adaptive", "closed loop", "deep brain stimulation", "neuromodulation", and combinations thereof were searched. The results were combined with articles from relevant references within the retrieved articles and review articles. Original research studies on adaptive deep brain stimulation in patients with Parkinson's disease were included.

for adequate motor symptom suppression.[18] Another considerable side-effect is gait deterioration, which can be induced by either insufficient or excessive stimulation. Too little stimulation can result in bradykinetic gait, whereas too much stimulation can lead to dyskinetic or unstable gait.[19] Managing gait disturbances in patients undergoing DBS is complex due to the interplay between disease progression and ongoing dopaminergic medication regimens. Balancing the treatment of bradykinesia, rigidity, and tremor against the risk of dyskinesia is a further challenge. Although dyskinesias often resolve after the use of DBS if dopaminergic medication is reduced, some patients continue to have dyskinesias, despite optimal programming of DBS settings.[20] Lastly, non-motor side-effects of DBS, such as apathy,[21] might make treatment complex. Although the exact mechanism underlying DBS-induced apathy is unclear, the reduction of dopaminergic medication after the use of DBS is thought to play a role. An altered degree of impulsivity can also be observed after STN DBS,[22] but the direction of the effect seems to depend on the exact stimulation site, with ventral stimulation sites in the STN—functionally associated with limbic processing—being linked to changes in impulsivity.[23,24]

The fundamental stimulation algorithm has been largely unchanged since the introduction of DBS.[25] Strategies to optimise stimulation include the avoidance of current spread outside the dorsolateral part of the STN by selecting the stimulation contact based on postoperative imaging, applying directional stimulation, or by adjusting stimulation settings, such as amplitude, frequency, and pulse width. These attempts to, for example, alleviate speech disturbances without compromising motor symptom control, are at best modestly successful.[26] Although stimulation-induced side-effects are heterogeneous and their causes often multifactorial, they typically seem to arise from an imbalance between insufficient or excessive stimulation and medication. This intricate relationship between symptom improvement and side-effect manifestation is well recognised, highlighting the paradoxical interplay between DBS and symptom severity. Specifically, patients with bradykinesia tend to benefit from DBS, whereas the motor function of those without substantial bradykinesia can worsen.[27,28] This paradox has led to the concept of DBS as having pros and cons—although pathological neural activity can be suppressed, excessive stimulation can also inhibit physiological activity, potentially leading to adverse effects. This understanding has contributed to the development of adaptive DBS (aDBS), which, in contrast to continuous DBS (cDBS), dynamically adjusts the stimulation in response to fluctuating pathological neural activity that might occur throughout the day. By keeping the neural signal features indicative of the physiological state (so-called physiomarkers) within a particular range, aDBS aims to maintain an optimal balance between symptom relief and the preservation of typical motor function by avoiding stimulation-related side effects.[29]

The concept of aDBS is founded on a closed-loop system that dynamically adjusts the stimulation in real-time, based on feedback signals from concurrent electrophysiological recordings from the same DBS electrodes or from electrocorticography, or based on accelerometer or gyroscope signals from inertial measurement units in wearable devices. Specific signal features from these recordings are then used as physiomarkers of symptom presence or severity. When these signals are detected by a computer algorithm, the stimulation settings are automatically adjusted without human interference. For example, stimulation can be switched on when the amount of recorded pathological beta oscillations exceeds a predefined threshold and switched off during periods with low beta oscillations. A related form of adaptive neuromodulation therapy has already been in use for several years in patients with epilepsy who do not respond to anti-seizure medication and for whom resection of epileptogenic brain tissue is not an option. Upon the detection of epileptiform activity by a fully implanted system, short electrical pulses are automatically delivered to normalise the neural activity.[30] Aside from reducing seizure rates, the therapy can aid in chronotype classification with the collection of neurophysiological data, which in turn can be used to adjust the dosing and timing of medication.[31] For treatment of Parkinson's disease, the greatest potential of aDBS lies in delivering a personalised and responsive approach to managing symptom variability, while minimising stimulation-induced side-effects. The following sections explain the mechanisms of aDBS in more detail, show clinical evidence supporting its efficacy, discuss the technical and practical challenges it presents, and describe future directions for its development.

Mechanisms and physiomarkers for aDBS in Parkinson's disease

Subcortical electrophysiology

Recording subcortical neural signals has been an area of interest for more than 60 years. The introduction of DBS and the temporary externalisation of electrode leads in humans has enabled the recording of local field potentials (LFPs) outside the operating theatre. The initial goal of such recordings was to study the response of subcortical neural activity to stimulation and medical treatment in Parkinson's disease, with the first study done by Brown and colleagues in 2001.[32] This study showed the presence of a spectral peak in the beta frequency range in the OFF dopaminergic state, including the synchronisation of beta activity between the STN and the GPi. After levodopa intake, beta activity decreased and gamma activity (60–90 Hz) increased. Subsequent studies showed that switching on DBS exerts similar effects to levodopa intake. To date, studies have indicated statistically significant associations between beta-based LFP measures, contralateral akinesia, and rigidity severity.[33] Thus, spectral power (often represented by power spectral density estimates) in the beta frequency range has emerged as the primary physiomarker for diagnostics and informing DBS treatment response in Parkinson's disease (figure 1).

Beta power is typically estimated across a defined time window of LFP signals, in which the medication state of the patient is assumed to be stable. However, in the time domain, beta oscillations are not sustained, but occur intermittently in brief episodes of a few hundred milliseconds, called bursts. The volatility of beta oscillations and their relation to spectral power was first shown in non-human primates by Feingold and colleagues in 2015.[37] In 2017, Tinkhauser and colleagues[38] showed that beta power increased together with beta burst duration in relation to symptom burden. The authors also observed that aDBS selectively suppressed the pathological protracted bursts, whereas cDBS suppressed all beta activity. The authors also showed that, specifically, the percentage of long, not short, beta bursts in a given interval is positively correlated with clinical impairment, and that the occurrence of longer bursts is reduced with levodopa medication.[39] A large study of 106 patients with Parkinson's disease showed that the mean beta burst duration during the OFF dopaminergic state was around 600 ms, and around 350 ms in the ON dopaminergic state.[40] Therefore, sensing duration cutoffs within aDBS algorithms might be clinically relevant, especially in algorithms with sub-second alternating stimulation amperages.[41] Furthermore, the low beta range (~13–20 Hz) within the beta

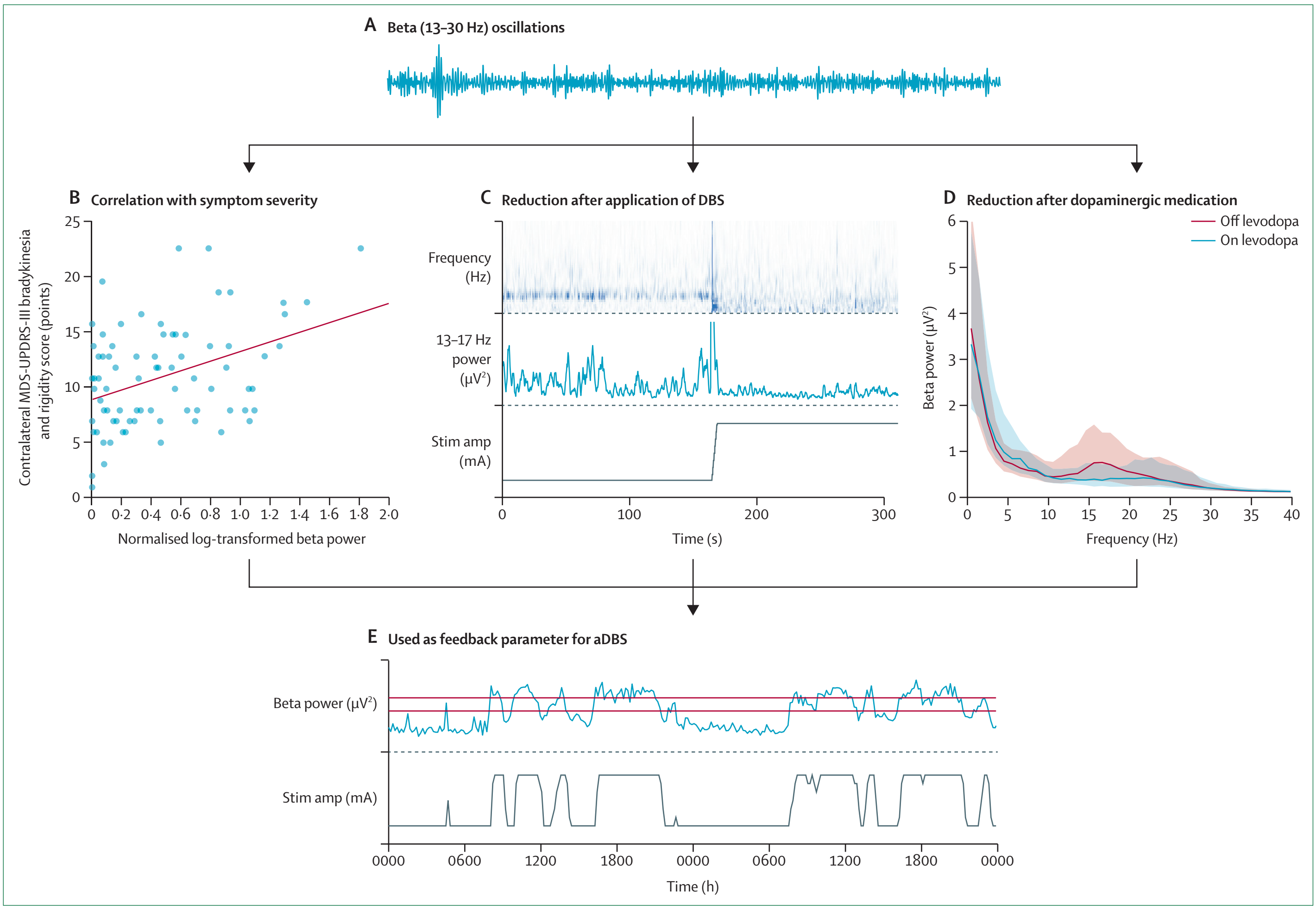

Figure 1: **Mechanism of action for beta-informed aDBS**

(A) LFP recordings from the subthalamic nucleus or internal segment of the globus pallidus show time-varying fluctuations in the amplitude of beta band (13–30 Hz) neural oscillations. (B) Adapted from Beudel et al,[34] by permission of John Wiley and Sons. In patients, the average amplitude of beta oscillations is correlated with the severity of contralateral bradykinesia and rigidity, as measured with the MDS-UPDRS-III. (C) Adapted from Swinnen et al.[35] Amplitude of beta oscillations decreases after the application of DBS (bottom), as observed from the time-frequency spectrum of the LFP recording (top) or the extracted beta amplitude envelope (middle) in this example case. (D) Amplitude of beta oscillations decreases after the intake of dopaminergic medication, as seen in the average power spectra from LFP recordings obtained in 52 patients (100 hemispheres). Unpublished data from a prospective observational cohort study.[36] (E) Excerpt of data from a patient treated with aDBS in our centre. aDBS automatically adjusts the stimulation amplitude (bottom) based on the signal strength of a specific physiomarker, in this case beta power (top). Red horizontal lines represent the lower and upper thresholds of the dual threshold aDBS control algorithm. The slight mismatch between the threshold crossings of beta power and the corresponding stimulation amplitude adjustments arises due to a difference in time resolution at which the beta power estimates are stored (10-min averages) compared with what is used by the control algorithm (s to min). aDBS=adaptive deep brain stimulation. DBS=deep brain stimulation. LFP=local field potential. MDS-UPDRS-III=Movement Disorder Society Unified Parkinson's Disease Rating Scale part III.

band potentially plays a different physiological role than the high beta range (~21–35 Hz), and responds differently to both dopamine and DBS.[42] That is, low beta range peaks respond to both dopamine and DBS, whereas peaks in the high beta range only respond to DBS.[43,44]

Most studies investigated beta band responses to dopaminergic medication or stimulation in isolation. Therefore, little is known about LFP fluctuations under the simultaneous influence of dopaminergic medication and stimulation, and such knowledge is important for the clinical implementation of aDBS algorithms. Observations have shown that finely tuned gamma activity, also referred to as entrained gamma, as a physiomarker of the levodopa ON state, can shift in frequency from within the natural 60–90 Hz range to exactly half the stimulation frequency (62·5 Hz with 125 Hz stimulation). This occurrence might be more indicative of the medication and symptom state than beta power.[45] Future research should establish the physiomarker most suitable for aDBS at the individual patient level.

Choice of aDBS algorithm

Currently, two types of aDBS algorithm are available, which differ mainly in their temporal resolution: single-threshold and dual-threshold (figure 2). Single-threshold algorithms respond to sub-second beta synchronisation bursts[39,41] by using one threshold value for detecting pathological beta oscillations. Dual-threshold algorithms respond to longer beta activity trends—eg, induced by levodopa intake[44]—leaving a buffer zone between an upper (related to the OFF state) and lower (related to the ON state) LFP threshold. Therefore, single-threshold algorithms can target the pathophysiology of Parkinson's disease more directly than dual-threshold algorithms by acting on the pathological burst activity.[39] However, dual-threshold algorithms are less prone to artifact-induced false detections of pathological activity, due to the longer time windows that are used for estimating the physiomarker, resulting in more extensive feedback signal smoothening. Whether a specific combination of DBS algorithm and physiomarker is effective for most patients with Parkinson's disease, or more personalised algorithms based on symptom phenotype are needed, is still unclear.

Setting up aDBS

Regardless of the type of algorithm, the aDBS device should have access to a feedback signal that reflects the current state of the patient. In practice, estimates of beta power as a physiomarker are either obtained by signal processing of recorded LFP time series,[41] or are directly computed by the implanted device itself (on-chip) in the latest sensing-enabled aDBS devices.[44] These measurements can also help to guide the selection of the optimal stimulation contact, as contacts with the highest beta power (beta peak) often yield the best clinical response.[46,47] However, the feedback signal can only be obtained from the contacts surrounding the stimulation contact, which can be situated at a larger distance from the neuronal source of beta oscillations, potentially resulting in reduced signal amplitude. Although the behaviour of beta power in response to dopamine and DBS is well studied, how LFP signals are modulated in response to aDBS and how this modulation affects aDBS performance are unclear. Therefore, beta power thresholds might have to be adjusted over time to optimise aDBS settings

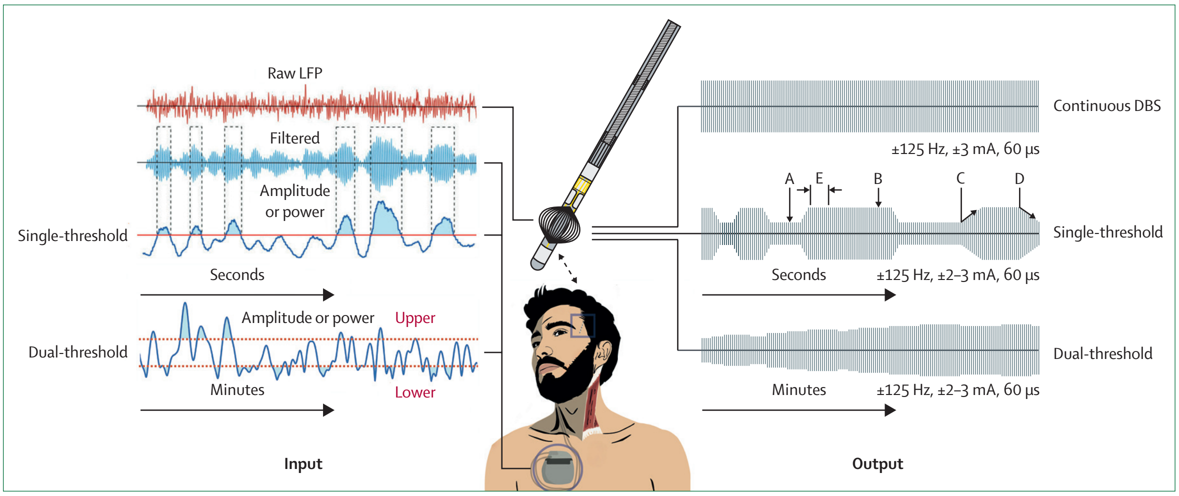

Figure 2: **Schematic of the workings of aDBS**
In the centre, a patient is depicted with a fully implanted DBS system consisting of electrodes in the subthalamic nucleus and a pulse generator. A magnified DBS electrode is depicted above (real size=1·57 mm in diameter). LFP recordings (in red) are used by the pulse generator to extract a physiomarker. The continuously measured physiomarker acts as feedback signal to control the stimulation. The two different types of aDBS control algorithms—single-threshold and dual-threshold—are depicted on the left. The stimulation amplitude (depicted on the right) changes between a low (A) and high (B) value following a transition time up (C) and down (D), over a period of seconds. A refractory (blanking) period (E) can be chosen, during which the stimulation amplitude cannot change. For single-threshold aDBS, stimulation is applied on either the low (A) or high (B) amplitude values, whereas for dual-threshold aDBS, the stimulation amplitude transitions more gradually between these two limits. aDBS=adaptive deep brain stimulation. DBS=deep brain stimulation. LFP=local field potential.

according to how the physiomarker fluctuates. Since the choice of physiomarker, LFP thresholds, blanking duration, ramping durations, and onset times of stimulation can all influence aDBS efficacy (figure 2), adjusting these settings should be considered when the aDBS effect is insufficient.

Clinical evidence for aDBS in Parkinson's disease

At present, evidence for the efficacy of aDBS for Parkinson's disease almost solely stems from investigational device studies. Since the publication of the first aDBS study[41] in humans in 2013, 28 clinical studies on aDBS have been published, which collectively included a total of 248 unique patients (148 male, 66 female, and 34 of unknown sex or gender) with Parkinson's disease (table). The STN was targeted in all single-target patients undergoing DBS, except in one case report describing a patient undergoing single-target GPi-DBS.[70] Dual-target aDBS was used in two studies, in which GPi-DBS was combined with either STN[73] or pedunculopontine nucleus DBS.[71] A clear shift from the perioperative[41,48–54,57,60,61,65] to the more chronic setting of weeks to years after implantation[45,55,56,58,59,62–64,66–73] has been observed. Most studies used the beta amplitude from STN LFP recordings to inform the aDBS algorithm,[41,48–66] and, in four studies, the aDBS algorithm was informed by the gamma amplitude from STN LFP recordings[45] or by electrocorticography from the sensorimotor or motor

	Setting	Feedback signal	Control algorithm	Patients undergoing aDBS (N; male/female)	Clinical outcome measure	aDBS compared with no stimulation	aDBS compared with cDBS
Single-target STN-DBS, STN beta power-informed							
Little et al, 2013[41]	Acute postoperative	Subthalamic 13–35 Hz peak frequency amplitude (bandwidth 2–6 Hz); one hemisphere	Single-threshold; stimulation either OFF or ON	8 (unknown)	MDS-UPDRS-III hemibody	49·7% improvement	More improvement (27·0% [SD 7·8]) for aDBS
Rosa et al, 2015[48]	Acute postoperative	Subthalamic 13–17 Hz band power	No threshold; linear adaptation between 0 V and 2 V	1 (1/0)	MDS-UPDRS-III upper limb hemibody; Rush-DRS	NA	Lower MDS-UPDRS-III scores for aDBS; no statistical difference in dyskinesia scores; no numerical outcome values reported
Little et al, 2016[49]	Acute postoperative	Amplitude of subthalamic activity with an individualised frequency and bandwidth encompassing peaks within 13–30 Hz	Single-threshold; stimulation either OFF or ON	4 (4/0)	MDS-UPDRS-III full motor examination	43% improvement	NA
Little et al, 2016[50]	Acute postoperative	Subthalamic 13–35 Hz peak frequency amplitude; one hemisphere	Single-threshold; stimulation either OFF or ON	8 (unknown)	MDS-UPDRS-III excluding rigidity; SIT	No difference in speech intelligibility (67·9% [SD 9·2] *vs* 70·4% [SD 6·4])	Lower MDS-UPDRS-III scores (19·7 points [SD 1·0] *vs* 31·6 points [SD 4·3]) and better speech intelligibility (70·4% [SD 6·4] *vs* 60·5% [SD 8·2]) for aDBS
Rosa et al, 2017[51]	Acute postoperative	Power of frequency window within subthalamic 13–30 Hz band; selection method unclear	Single-threshold; stimulation either OFF or ON	10 (unknown)	MDS-UPDRS-III; UDysRS	NA	Similar MDS-UPDRS-III improvement ON medication (−46·1% [SD 10·5] *vs* −40·1% [SD 17·5]); lower dyskinesia scores (11·7 points [SD 6·7] *vs* 15·0 points [SD 8·7]) for aDBS
Arlotti et al, 2018[52]	Acute postoperative	Power of subthalamic 11–35 Hz peak frequency ±2 Hz	No threshold; linear adaptation between 0 V and clinically effective amplitude	11 (7/4)	MDS-UPDRS-III; UDysRS	Lower MDS-UPDRS-III score OFF medication (22·2 points [SD 3·3] *vs* 29·4 points [SD 3·9]) for aDBS and similar scores ON medication (17·9 points [SD 2·0] *vs* 15·5 points [SD 2·3]); no differences in dyskinesia scores in both OFF and ON medication	NA
Herz et al, 2018[53]	Acute postoperative	Power of subthalamic 13–30 Hz peak frequency	Single-threshold; stimulation either OFF or ON, with 250 ms ramping	7 (unknown)	MDS-UPDRS-III; response time during decision making task	Reduction in response time (−8·8%) but not accuracy for aDBS	No difference in MDS-UPDRS-III score OFF medication (22·2% *vs* 22·9%); no difference in response time or accuracy

(Table continues on next page)

	Setting	Feedback signal	Control algorithm	Patients undergoing aDBS (N; male/female)	Clinical outcome measure	aDBS compared with no stimulation	aDBS compared with cDBS
(Continued from previous page)							
Velisar et al, 2019[54]	Acute postoperative	Subthalamic 13–30 Hz band power	Dual-threshold; adaptation between 0 V and clinically effective amplitude	13 (11/2)	Kinematic measures of bradykinesia and tremor (only tremor-dominant patients)	Improvement of bradykinesia for aDBS; no numerical outcome values reported; median time below tremor detection threshold 95·4% (95% CI 65·4–100·0)	NA
Piña-Fuentes et al, 2020[55]	Chronically implanted during battery replacement	Power of subthalamic 12–35 Hz peak frequency ±3 Hz	Single-threshold; stimulation either OFF or ON	13 (11/2)	MDS-UPDRS-III upper limb bradykinesia and tremor; SIT; tablet-based bradykinesia test	Lower MDS-UPDRS-III subscore (12·30 points [SD 0·85] *vs* 15·81 points [SD 0·75]) for aDBS; no difference in speech intelligibility or bradykinesia	No difference in MDS-UPDRS-III score or bradykinesia
Petrucci et al, 2020[56]	Unknown	Subthalamic 12–18 Hz band power	Dual-threshold; adaptation between 0 V and clinically effective amplitude; different for up and down	1 (1/0)	Percentage freezing during stepping in place task	Less freezing OFF medication (5·2% *vs* 54·9%)	Similar freezing with aDBS compared with clinical cDBS (5·2% *vs* 2·3%) but lower compared with cDBS matched to aDBS in contact and settings (5·2% *vs* 23·5%)
Bocci et al, 2021[57]	Acute postoperative	Power of subthalamic 11–35 Hz peak frequency ±2 Hz	No threshold; linear adaptation within therapeutic window	8 (6/2)	MDS-UPDRS-III relative score; MDS-UPDRS-III subscores for rest tremor, bradykinesia, and rigidity; Rush-DRS	NA	Lower MDS-UPDRS-III relative score (0·33 [SD 0·04] *vs* 0·46 points [SD 0·05]) for aDBS; lower MDS-UPDRS-III rigidity subscore (2·14 points [SD 0·50] *vs* 2·91 points [SD 0·66]) for aDBS but no differences in rest tremor and bradykinesia subscores; lower dyskinesia scores (1·67 points [SD 0·53] *vs* 2·50 points [SD 1·13]) for aDBS
Gilron et al, 2021[58]	Chronic	Power of subthalamic 12–30 Hz peak frequency ±4 Hz	Dual-threshold; adaptation within therapeutic window	1 (0/1)	No clinical outcomes for STN-aDBS	NA	NA
Nakajima et al, 2021[59]	Chronic	Power of subthalamic 16·60 Hz ±2·5 Hz; selection method unclear	Dual-threshold; adaptation within therapeutic window	1 (1/0)	MDS-UPDRS-IV parts A (dyskinesias) and B (motor fluctuations)	NA	No difference in dyskinesia scores (0 points vs 0 points); lower motor fluctuation scores (0 points *vs* 3 points) for aDBS
He et al, 2023[60]	Acute postoperative	Power of subthalamic 13–30 Hz peak frequency ±3 Hz	Single-threshold; adaptation between 0 mA and clinically effective amplitude	13 (7/6)	Reaching task; finger tapping movements with blinded bradykinesia ratings; tremor assessment with accelerometer; estimation of effect β with GLME model	Improvement of reaction time (β=–0·0253 [SD 0·0094]) and velocity (β=0·0128 [SD 0·0045]) during reaching task; increased acceleration during finger tapping (β=0·0339 [SD 0·0149]) and reduction of blinded bradykinesia ratings (β=–0·1738 [SD 0·0593]) and lower resting tremor (β=–0·5726 [SD 0·1256])	No difference in task performance metrics and bradykinesia, but more resting tremor (k=0·7605 [SD 0·2179]) for aDBS
Busch et al, 2024[61]	Acute postoperative	Subthalamic 12–30 Hz band power	Single-threshold; adaptation between 0 V and clinically effective amplitude	12 (8/4)	No clinical outcome measure	NA	NA
Patel et al, 2025[62]	Chronic	Subthalamic beta power; details unknown	Unknown	1 (1/0)	Complete MDS-UPDRS; PDQ-39	NA	Reduction in complete MDS-UPDRS score (70 points to 50 points) and MDS-UPDRS-III score (42 points to 28 points); increase in PDQ-39 score (no numerical outcome given)

(Table continues on next page)

	Setting	Feedback signal	Control algorithm	Patients undergoing aDBS (N; male/female)	Clinical outcome measure	aDBS compared with no stimulation	aDBS compared with cDBS
(Continued from previous page)							
Busch et al, 2025[63]	Chronic	Power of subthalamic 13–30 Hz peak frequency ±2·5 Hz	Dual-threshold; adaptation between two clinically relevant stimulation amplitudes	8 (5/3)	Ecological momentary assessments (7-point scales) on overall wellbeing, general movement, tremor, dyskinesias, and clinical status	NA	Improvement in overall wellbeing (5·92 points [SD 1·01] to 6·73 points [SD 1·33]); no significant differences in motor symptoms and clinical status; only group-level results reported
Emura et al, 2025[64]	Chronic	Power of subthalamic 13–30 Hz peak frequency ±2·5 Hz	Dual-threshold; adaptation between two clinically relevant stimulation amplitudes	19 (8/11)	MDS-UPDRS-I; MDS-UPDRS-II; MDS-UPDRS-III in four conditions; MDS-UPDRS-IV; PDQ-39; Berg Balance Scale; 10-minute timed up and go	Reduction in MDS-UPDRS-III scores without medication at 12 months (26 points [range 23–32] *vs* 53 points [range 42–57]); not statistically compared	Reduction in MDS-UPDRS-III score OFF medication and ON stimulation (baseline 33 points [range 28–38] to 26 points [range 23–32] at 12 months); no differences in other outcomes but correlation between age at initiation and PDQ-39 improvement at 12 months (r=0·616)
Ferrucci et al, 2025[65]	Acute postoperative	Power of subthalamic 11–35 Hz peak frequency ±2 Hz	No threshold; linear adaptation within stimulation window; limits unclear	16 (12/4)	Cognitive assessments of attention, language, and memory	NA	Fluctuations in reaction time between OFF and ON medication conditions during cDBS (n=4) but not during aDBS (n=16); other cognitive outcomes did not fluctuate between medication conditions
Bronte-Stewart et al, 2025[66]	Chronic	Power of subthalamic 8–30 Hz peak frequency ±2·5 Hz	Single-threshold or dual-threshold; adaptation between two clinically relevant stimulation amplitudes	68 (48/20)	Self-reported home diary; CGI-C; MDS-UPDRS-I; MDS-UPDRS-II; MDS-UPDRS-III; MDS-UPDRS-IV; PDQ-39	NA	Clinically meaningful increase in ON time without troublesome dyskinesias for dual-threshold (+1·3 h [SD 0·5]) but not for single-threshold (+0·6 h [SD 0·6]); clinically meaningful reduction in MDS-UPDRS-III score (–3·73 points [SD 1·17]) and UPDRS-IV (–1·6 points [SD 3·32]) scores on the selected aDBS mode
Single-target STN-DBS, gamma-informed							
Swann et al, 2018[67]	Chronic	Power of cortical 80 Hz ±1·25 Hz (half the stimulation frequency)	Single-threshold; adaptation within therapeutic window	2 (2/0)	MDS-UPDRS-III upper body bradykinesia; UDysRS	NA	Only one patient with clinical outcomes; similar bradykinesia and dyskinesia scores; no statistical tests done
Gilron et al, 2021[58]	Chronic	Power of cortical 50–90 Hz peak frequency ±4 Hz	Dual-threshold; linear adaptation within therapeutic window	1 (0/1)	Kinematic data and patient diary	NA	Increase in ON time during aDBS; no statistical tests done
Oehrn et al, 2024[45]	Chronic	Cortical 64–66 Hz or 64–69 Hz band power (n=3) or subthalamic 64–66 Hz band power (n=1)	Single-threshold (n=3) and dual-threshold (n=1); adaptation within therapeutic window	4 (4/0)	Daily questionnaire on symptoms and quality of life (EQ-5D-5L); estimation of effect β with GLME model	NA	Less time with most bothersome symptom (β=–16·3% [SD 4·4]) without worsening time with opposite symptom (β=–2·5% [SD 2·2]) and higher quality of life (β=6·9% [SD 1·7]) for aDBS
Cernera et al, 2025[68]	Chronic; 8 month follow-up of one patient from Oehrn et al, 2024[45]	Cortical 64–66 Hz band power	Dual-threshold (n=1); adaptation within therapeutic window	1 (1/0)	Daily questionnaire on symptoms and quality of life (EQ-5D-5L); bradykinesia and dyskinesia severity with a wearable device; estimation of effect β with GLME model	NA	Less time with bradykinesia (β=–10·22% [SD 1·67]) for aDBS without worsening time with dyskinesia (β=0·26% [SD 0·29]); higher quality of life (β=10·38% [SD 1·37]) for aDBS; wearable device outcomes: reduction of fluctuations in bradykinesia (β=–8·66% [SD 1·74]; p<0·001) for aDBS

(Table continues on next page)

	Setting	Feedback signal	Control algorithm	Patients undergoing aDBS (N; male/female)	Clinical outcome measure	aDBS compared with no stimulation	aDBS compared with cDBS
(Continued from previous page)							
Other configurations							
Malekmohammadi et al, 2016[69]	Chronic (STN)	4–8 Hz tremor power measured with a wearable device	Dual-threshold; adaptation between 0 V and highest tolerable amplitude	5 (3/2)	4–8 Hz tremor power measured with a wearable device	Improvement of 36·6% with aDBS	NA
Piña-Fuentes et al, 2019[70]	Chronically implanted during battery replacement (GPi)	Pallidal 18 Hz ±3 Hz band power	Single-threshold; stimulation either OFF or ON	1 (0/1)	MDS-UPDRS-III upper body	Improvement of 64% (4 points *vs* 11 points) for aDBS	Greater improvement (64% *vs* 37%; scores of 4 points *vs* 7 points compared with 11 points without stimulation) for aDBS
Molina et al, 2021[71]	Chronic (GPi and PPN)	2·5–7·5 Hz band power within PPN	Single-threshold; stimulation either OFF or ON for PPN lead	5 (3/2)	Number of FoG episodes; gait performance with three dimensional motion capture	>40% improvement in 3 of 5 patients (average 29% [range −63% to 93%]) for aDBS; no difference at group level; no change in velocities and stride lengths at group level	NA
Smyth et al, 2023[72]	Chronic (STN)	Cortical delta (0·5–4·5 Hz) and beta (12–30 Hz) power	Linear discriminant analysis model for sleep stage classification	1 (unknown)	No clinical outcome measure	NA	NA
Schmidt et al, 2024[73]	Chronic (STN and GPi)	Subthalamic 13–30 Hz band power	Proportional-integral controller; adaptation between 0 mA and cDBS amplitude	6 (4/2)	MDS-UPDRS-III ON and OFF medication (in-clinic); dyskinesia and tremor measured with wearable device (at home, n=3)	NA	Similar improvement in MDS-UPDRS-III scores both ON (mean difference=−1) and OFF medication (mean difference=0·25); at-home: similar reduction in tremor; dyskinesias not routinely observed

When available, outcomes of aDBS compared with no stimulation or with cDBS are provided and reported as the mean (SD) unless indicated otherwise. CGI-C=Clinical Clobal Impression of Change Scale. DBS=deep brain stimulation. aDBS=adaptive deep brain stimulation. cDBS=continuous deep brain stimulation. EQ-5D-5L=EuroQol Five-Dimensional, Five-Level Questionnaire. FoG=freezing of gait. GLME=generalised linear mixed-effects. GPi=internal segment of the globus pallidus. MDS-UPDRS=Movement Disorder Society Unified Parkinson's Disease Rating Scale. NA=not applicable. PDQ-39=39-item Parkinson's Disease Questionnaire. PPN=pendunculopontine nucleus. Rush-DRS=Rush Dyskinesia Rating Scale. SIT=speech intelligibility test. STN=subthalamic nucleus. UDysRS=Unified Dyskinesia Rating Scale.

Table: **Literature overview of aDBS in Parkinson's disease**

cortex.[45,58,67,68] The feedback signals in the remaining studies were beta power in the GPi,[70,72] 1–8 Hz activity in the pedunculopontine nucleus,[71] both delta (0·5–4·5 Hz) and beta amplitude recorded from the primary motor cortex,[72] and the amplitude of resting tremor measured with a wearable device.[67] Despite similarities in physiomarkers, the control algorithms for aDBS substantially differed between studies (table). For the sake of relevance to current clinical practice, we first focus on results from studies with single target STN DBS and beta activity as a physiomarker.

The first proof-of-concept study in 2013 included eight patients with bilaterally implanted electrodes in the STN who received cDBS, aDBS, or random bursts of stimulation.[41] During aDBS, the LFP amplitude around the beta peak frequency was used to switch the stimulation either off or on at a clinically effective stimulation amplitude that did not result in side-effects. Testing was done within the acute postoperative period in blocks of roughly 10 min in a randomised order. Compared with no stimulation, both cDBS (30·5%) and aDBS (49·7%, based on blinded assessment; p=0·007) yielded a significant improvement in a composite score of bradykinesia, rigidity, and tremor, whereas random stimulation did not. The increased motor improvement during aDBS compared with cDBS was also statistically significant (27·0% [SD 7·8]; p=0·005).

Since then, other studies have replicated this finding: compared with no stimulation and without dopaminergic medication, beta-informed aDBS reduces scores of bradykinesia, rigidity, and tremor as measured with kinematics,[54] the full motor examination of the Movement Disorder Society Unified Parkinson's Disease Rating Scale (MDS-UPDRS-III),[48,52,57,64] which also includes axial symptoms such as postural instability,[74] or measured with subsets of MDS-UPDRS-III items.[55] One study comparing aDBS with no stimulation found a positive effect of aDBS on motor symptoms when patients were off dopaminergic medication but no significant difference when patients were on medication.[52] Across studies, the reported reduction in full MDS-UPDRS-III motor examination scores with aDBS compared with no stimulation ranges from 27·2% to 50·9%.[41,51,52,64]

When directly comparing the effects of aDBS with cDBS on motor symptoms, most studies have found no significant difference,[51,53,60,63,66] a superior effect of aDBS over cDBS,[48,50,57,62,64,66] or only a positive effect on bradykinesia.[55] However, one study reported a lower reduction of resting tremor during aDBS than with cDBS.[60] In the ADAPT-PD trial, 95% (dual-threshold aDBS) and 87% (single-threshold aDBS) of participants had similar ON time without troublesome dyskinesias compared with cDBS.[66] In addition, an increase in ON time without troublesome dyskinesias (+1·3 h [SE 0·5]) and an improvement of MDS-UPDRS-III score (–3·73 points [SD 1·17]) were seen during dual-threshold-aDBS compared with cDBS. The effect of beta-informed aDBS on freezing of gait was specifically assessed in one case report, in which aDBS did better in reducing the percentage of time freezing (1·5%) compared with cDBS (23·5% *vs* 68·7% without stimulation) when matched to aDBS in contacts and settings but not better than cDBS as clinically applied (2·3%).[55] The clinically used cDBS contacts of this patient were not sensing-compatible and, therefore, could not be used for aDBS. Overall, studies report substantially lower time on stimulation with beta-informed aDBS (ranging from 42·6% to 48·8%) than with cDBS.[41,49,55,60]Similarly, aDBS was shown to deliver less total electrical energy to the surrounding tissue (reductions ranging from 13% to 73·6%),[41,49,51,54,57,66] although one study reported either no difference or only a 2% reduction in total electrical energy delivered for the two aDBS configurations used.[56]

Evidence seems to suggest that, within the experimental setting, beta-informed aDBS appears to be effective in reducing the motor symptoms of Parkinson's disease to an extent at least similar to cDBS, although a superior effect over cDBS cannot be concluded with certainty. However, as mentioned earlier, the greatest potential of aDBS in Parkinson's disease lies not necessarily in increased optimal control of motor symptoms, but in maintaining symptom control while minimising stimulation-induced side-effects. Beta-informed aDBS resulted in a reduction of dyskinesias,[48,51,57] motor fluctuations,[59] and motor complications[66] compared with cDBS. The additive effect of aDBS on levodopa did not elicit more dyskinesias than levodopa alone.[52] Furthermore, two studies with a total of 23 patients showed that, on a group level, cDBS induced dysarthria, whereas aDBS did not.[50,55] To date, no studies have assessed the effect of aDBS on other stimulation-induced side-effects, such as gait deterioration, apathy, or impulsivity. In addition, no studies have assessed the effects of aDBS on non-motor symptoms, such as depression, sleep quality, or autonomic dysfunction. Only two studies have been published on cognitive effects, indicating stable cognitive performance under aDBS in various medication states,[65] and similar effects of aDBS and cDBS on decision making.[53] The effects of aDBS on sleep quality or sleep–wake disorders should be carefully assessed since the typical decrease in beta activity during sleep[75] might result in a decrease in stimulation intensity.

Of the remaining aDBS studies, four reported clinical results from gamma-informed STN aDBS.[45,58,67,68] Seven different patients were described, all chronically implanted with subthalamic leads. The aDBS algorithms made use of gamma activity recorded with subthalamic LFPs[45] or electrocorticography from the motor cortex[58,67] or the broader primary sensorimotor cortex.[45,68] Contrary to beta power, gamma power shows an inverse relation between signal amplitude and motor symptom severity; therefore, stimulation amplitude decreases in gamma-informed aDBS following an increase of the feedback signal. In most patients, a narrow bandwidth (2–5 Hz) bandpass filter was used, which encompassed half the stimulation frequency.[45,67,68] Two studies, each describing one patient with gamma-informed aDBS and clinical outcomes, reported an increase in ON time without dyskinesias[58] or similar scores in bradykinesia and rigidity[67] compared with cDBS. More substantial evidence can be found in a study comparing cDBS with aDBS at home in blinded randomised blocks of 2–7 days within 1 month. The authors found that aDBS decreased the amount of time patients spent with their most bothersome symptoms, without increasing the time with the opposite symptom (eg, bradykinesia *vs* dyskinesia).[45] In addition, aDBS resulted in a higher quality of life than cDBS in three of the four patients. Long-term data from one of these patients, in which cortical gamma activity was recorded, show that the beneficial effects on bradykinesia and quality of life was sustained over 8 months.[68] Although limited in patient number, these results show that aDBS informed by finely tuned gamma also holds promise in treating the motor symptoms of patients with Parkinson's disease.

Technical and practical challenges

Despite the promise of aDBS, its widespread implementation faces several technical and practical challenges, including physiomarker variability, hardware limitations, and the need for standardised protocols and clinician training. Current European conformity-marked (CE) aDBS systems include Newronika's AlphaDBS and Medtronic's Percept device with BrainSense software. Although AlphaDBS has only received investigational device exemption from the US Food and Drug Administration, the BrainSense aDBS software has received full approval. These systems can only modulate the stimulation amplitude based on beta power in the case of AlphaDBS, and based on both alpha (8–12 Hz) and beta power in the case of the Percept device. Both systems require a stable frequency of interest that reliably reflects the presence and severity of Parkinson's disease symptoms. Findings indicate that, after initial fluctuations within

3 months after DBS implantation, the amplitude and frequency of the subthalamic beta peaks of patients with Parkinson's disease seem to be stable over time.[76,77] The stability of the beta peak amplitude was also shown in a cohort of patients who had been receiving DBS treatment for several years, while stimulation was on.[78] Notably, however, within the first 18 months after DBS implantation, frequency shifts (>5 Hz) and the disappearance of beta peaks occurred in 12–58% of patients, depending on the follow-up period and the peak selection method.[76] Hence, in some patients, adjustments of aDBS settings and configuration might be needed after initial programming and perhaps even on a regular basis.

In addition, although one or more beta peaks are discernible in LFP signals recorded at rest and during OFF medication in most hemispheres (>80%), for some patients, such peaks are absent.[79] This absence might be due to suboptimal electrode placement relative to the clinically effective site, as beta power is strongest in the dorsolateral part of the STN and the posteroventral GPi.[80,81] Moreover, the LFP recording montage might hinder beta peak detection, as the bipolar configuration can theoretically attenuate widespread synchronous activity with common noise rejection and through the orientation of neurons relative to the contacts used for recording.[82] If no clear spectral peak is observed, the stimulation amplitude can be modulated by the beta power recorded from the contralateral hemisphere, assuming a preserved interhemispheric correlation in beta dynamics and clinical symptom severity.

By contrast, multiple spectral peaks are observed in the beta range in some patients,[83,84] raising the question of which peak to prioritise as a physiomarker for aDBS control. The selection might depend on the clinical rationale—ie, whether a medication-responsive dual-threshold or a burst-responsive single-threshold algorithm is intended. Nonetheless, inter-individual differences in beta power were found to capture only approximately 17% of variability in bradykinesia and rigidity severity in patients,[33] and are thus not fully indicative of symptoms. Beta power shows short-term variability intra-individually as it comodulates with various behavioural states, such as movement of the limbs or fingers,[85] gait,[86] speech,[87] motor imagery,[88] action observation,[89] cognitive actions such as counting,[90] and sleep.[91] Not all states might require a change in stimulation; however, the states are not recognised by current aDBS algorithms.

The fully implanted nature of clinical aDBS systems imposes constraints on signal acquisition and quality, processing capacity, and data storage, requiring manufacturers to compromise between computational power, storage, battery longevity, and algorithmic complexity. Sensing-enabled DBS systems are susceptible to artifacts from movement,[91,92] cardiac activity,[93] abrupt stimulation shifts,[94] and stimulation itself, making physiomarker reliability complex. Current clinically available aDBS systems are limited to LFP-based control, without the option to integrate alternative physiological signals. Beta power within a 2·5 Hz range surrounding the patient-specific frequency of interest is calculated, but is stored only as 10-minute averages with a 60-day retention period. Only investigational devices offer expanded capabilities for aDBS refinement, such as high-resolution (eg, 256 Hz) broadband LFP export and alternative physiomarker-driven control strategies, including inverse modulation of stimulation amplitude, frequency-based modulation, or frequency of interest selection beyond alpha and beta bands.

Concurrent stimulation and sensing with the same electrode lead increases the susceptibility to stimulation artifacts. For aDBS, patients should have optimised DBS settings that permit LFP acquisition, obtained with a symmetric dipole (sandwiching) configuration around the stimulation contact, with monopolar stimulation referenced to the implanted neurostimulator (figure 2).[94] As a result, the technical limitations of currently available aDBS systems might restrict the ability to optimally tailor stimulation to the therapeutic needs of the patient.

With aDBS entering clinical practice, systematically integrating and disseminating theoretical, technical, and practical insights is crucial for standardised implementation. Establishing a consensus on patient selection criteria is a key step. Considering the amount of expertise and time required for aDBS programming due to its complexity, applying aDBS to all patients seems unfeasible at the present time. Moreover, aDBS will probably not be advantageous compared with cDBS for all patients. Benefitting patients might include those with refractory symptoms, despite optimised cDBS, or those with a suboptimal response due to stimulation-induced side-effects. A possible implementation approach would be to initially apply cDBS, after which conversion to aDBS could take place if needed. Beyond clinical considerations, aDBS necessitates a robust physiomarker that is indicative of the symptom burden. However, in-clinic physiomarker stability does not ensure reliability outside the hospital, due to differences in LFP recording modes, signal fluctuations induced by daily life activities,[59] and diurnal variations in beta power.[91] Validating the selected physiomarker and other settings as the best choice for the aDBS algorithm is challenging in practice, as evaluation relies on the subjective nature of patient-reported symptoms. At present, practical guidelines for the selection of aDBS stimulation settings are still absent.

Future directions

The last two decades have seen rapid technological advances in DBS electrode designs, battery efficiency, and pulse generator capabilities. The sheer amount of stimulation settings and temporal stimulation patterns now far exceed the understanding of how to reach optimal clinical outcomes. Adequate setting selection is already a challenge for cDBS, and the extra settings that come with aDBS add further complexity. Specifically, the choice of physiomarker, LFP thresholds, blanking duration, ramping durations, and onset times of stimulation can influence aDBS efficacy. Now that the first patients are being treated with aDBS in routine clinical care, systematic monitoring and data collection during the treatment process will help to identify suitable clinical workflows. However, the move from data collection to insights requires the combination of information from different centres. In addition, larger and better controlled trials are needed to verify the long-term benefits, safety, and cost-effectiveness of aDBS. Such trials should clarify what symptom domains of Parkinson's disease benefit most from aDBS and evaluate differences in overall programming time compared with cDBS. These studies will require careful design, including blinding, randomisation (eg, randomised controlled trials or crossover randomised controlled trials), and the selection of multidimensional outcome measures that cover functional domains (eg, activities of daily living), motor (eg, MDS-UPDRS-III and patient motor diary), and non-motor symptoms (eg, patient non-motor diary), and that consider battery longevity and time needed for aDBS programming. With more insights, targeted training programmes can be developed to equip clinicians with the expertise needed for effective patient selection, implementation, monitoring, and refinement, ensuring successful adoption in routine care.

The reliance of current aDBS algorithms on beta power as a single physiomarker of symptom severity might be, for some patients, insufficient to adequately capture the moments in time when stimulation should ideally be adjusted. Incorporating information from multiple physiomarkers—either from the same LFP signal used to extract beta power or from additional modalities, such as electrocorticography or wearable sensors—holds promise for further improving aDBS performance. These signals could be used to provide behavioural context, such as whether the patient is moving[95,96] or to monitor different symptoms in parallel. Finely tuned gamma activity in electrocorticography signals could indicate periods with dyskinesia,[45] and rhythmic patterns in accelerometer signals, measured with inertial measurement units integrated into a smartwatch, are predictive of tremor.[97] Adjustments in stimulation could then be implemented, based on a combination of features, to allow for more versatile control in different daily life situations and to better regulate side-effects. Alternatively, enhanced physiomarker detection accuracy for a specific symptom can be obtained by combining multiple signal features in a regression model.[98] Machine learning algorithms are expected to further improve the detection of symptoms by aiding the detection of optimal, and possibly non-linear, signal feature combinations,[99] which could be unique for individual patients. More sophisticated algorithms can also be implemented to accommodate time-dependent variations in threshold values and time window durations, acknowledging that the relationship between individual LFP markers and symptom occurrence might not be stable in time. Instead of manually altering aDBS settings during hospital visits, future algorithms might be capable of automatically refining the settings on a daily or weekly basis.

A further crucial step towards successful implementation of aDBS in clinical care is addressing the unique symptom profile, neurophysiological characteristics, medication schedules, daily life activities, and personal expectations of every individual patient. Instead of aiming for a universal solution, the individually tailored choice of physiomarkers and stimulation settings for aDBS algorithms will likely result in better outcomes. For example, physiomarkers might need to be identified based on individual so-called neuronal fingerprints, which could be variations or combinations of known physiomarkers, and stimulation thresholds might be programmed based on subjective experiences of side-effects. As the number of studies identifying associations between LFP signal features and different motor and non-motor symptoms continues to expand,[100] and knowledge regarding the underlying neurobiology of motor and cognitive functions and the effect of DBS continues to improve, other symptoms of Parkinson's disease, such as gait impairment, sleep disturbances, apathy, impulsivity, and depression, might also be effectively treated with aDBS. Ultimately, these progressive insights could lead to more personalised treatments that manage the multidimensional aspects of the disease.

Conclusion

Adaptive DBS represents an advancement in Parkinson's disease therapy by dynamically addressing motor fluctuations and reducing stimulation-induced side-effects. Its promise lies in integrating multiple physiomarkers, wearable sensors, and artificial intelligence-driven algorithms for more precise, personalised therapy. Future applications might extend to gait, sleep, and other non-motor symptoms. However, technical and clinical challenges remain—particularly in physiomarker selection and standardisation. Continued investment in robust clinical trials, real-world studies, and technology development is essential to establish the safety, efficacy, and accessibility of aDBS. As aDBS enters clinical practice, interdisciplinary collaboration will be key to translating innovation into long-term patient benefit.

Contributors

Conceptualisation: MB. Writing—original draft: all authors. Writing—review and editing: all authors. Supervision: MB.

Declaration of interests
MB received consulting fees and honoraria to their institution for educational events from Medtronic in 2023–2025. All other authors declare no competing interests.

Acknowledgments
MB received institutional grants from Amsterdam University Medical Center via the TKI-PPP programme (2021 and 2023), the Netherlands Organisation for Health Research and Development via the European Union Joint Programme—Neurodegenerative Disease Research, ParkinsonNL, and the stichting ParkinsonFonds. BCMvW received a grant from the Netherlands Organisation for Health Research and Development via the European Union Joint Programme—Neurodegenerative Disease Research.

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